

1. Roots of Equations

Bisection Method:

$$c = \frac{a+b}{2} \quad \text{where } f(a)f(b) < 0$$

Secant Method:

$$x_{n+1} = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$$

Newton–Raphson Method:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Fixed Point Iteration:

$$x_{n+1} = g(x_n) \quad \text{converges if } |g'(x)| < 1$$

2. Linear Algebraic Equations

Cramer's Rule:

$$x_i = \frac{\det(A_i)}{\det(A)}$$

Gauss–Jordan Elimination:

Transforms augmented matrix $[A|B]$ into reduced row echelon form $[I|X]$ using row operations, yielding $x_i = b'_i$.

Gauss-Jacobi Method:

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} a_{ij} x_j^{(k)} \right)$$

Gauss–Seidel Method:

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i+1}^n a_{ij} x_j^{(k)} \right)$$

Cholesky's Factorization Method:

Solve $AX = B$ where $X = [x, y, z]^T$, $B = [d_1, d_2, d_3]^T$.

Let $A = LL^T$:

$$\begin{bmatrix} a_{11} & b_{12} & c_{13} \\ a_{21} & b_{22} & c_{23} \\ a_{31} & b_{32} & c_{33} \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix}$$

Find l_{ij} . From $LL^T X = B$, put $L^T X = Y = [y_1, y_2, y_3]^T$.

Step 1: Solve $LY = B \rightarrow$ Find Y .

Step 2: Solve $L^T X = Y \rightarrow$ Find X .

3. Numerical Integration

Trapezoidal Rule: ($h = \frac{b-a}{n}$, $x_i = a + ih$)

$$\int_a^b f(x) dx \approx \frac{h}{2} [y_0 + 2(y_1 + y_2 + \dots + y_{n-1}) + y_n]$$

Simpson's 1/3 Rule:

$$\approx \frac{h}{3} [y_0 + y_n + 4 \sum \text{odd} + 2 \sum \text{even}]$$

Simpson's 3/8 Rule: (Requires multiple of 3 intervals)

$$\approx \frac{3h}{8} [y_0 + y_n + 3(y_1 + y_2 + y_4 + \dots) + 2(y_3 + y_6 + \dots)]$$

Distance Covered Math Problem: Calculate distance (S) from velocity (v) time-series data using integration: $S = \int v(t) dt$ (Apply Trapezoidal/Simpson's).

4. Ordinary Differential Equations

Euler's Method (to solve IVP):

$$y_{n+1} = y_n + hf(x_n, y_n)$$

Modified Euler Method + Formula:

$$\text{Predictor: } y_{n+1}^{(0)} = y_n + hf(x_n, y_n)$$

$$\text{Corrector: } y_{n+1}^{(k+1)} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1}^{(k)})]$$

Runge–Kutta Method (4th Order):

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = hf(x_n + h, y_n + k_3)$$

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Taylor Series Method:

$$y(x_0 + h) = y(x_0) + hy'(x_0) + \frac{h^2}{2!}y''(x_0) + \frac{h^3}{3!}y'''(x_0) + \frac{h^4}{4!}y''''(x_0) + \dots$$

Picard's Method: ($\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$)

$$y_n(x) = y_0 + \int_{x_0}^x f(t, y_{n-1}(t)) dt \quad ; n = 1, 2, 3, \dots$$

5. Error Analysis & Derivatives

Absolute and Relative Error:

$$E_a = |\text{True} - \text{Approximate}|$$

$$E_r = \left| \frac{\text{True} - \text{Approximate}}{\text{True}} \right|$$

Construct the Forward Difference Table:

$$\Delta y = y_{i+1} - y_i, \quad \Delta^2 y = \Delta y_{i+1} - \Delta y_i$$

First Derivative (dy/dx):

$$\frac{dy}{dx} = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \frac{1}{5} \Delta^5 y_0 \right]$$

Second Derivative (d^2y/dx^2):

$$\frac{d^2y}{dx^2} = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 \right]$$

6. Interpolation

What is Interpolation?

The process of estimating an unknown value that falls within a given sequence of discrete, known data points.

Lagrange Formula (Lagrange Interpolation):

$$f(x) = \frac{(x-x_1)\dots(x-x_n)}{(x_0-x_1)\dots(x_0-x_n)}f(x_0) + \frac{(x-x_0)(x-x_2)\dots(x-x_n)}{(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)}f(x_1) + \dots + \frac{(x-x_0)\dots(x-x_{n-1})}{(x_n-x_0)\dots(x_n-x_{n-1})}f(x_n)$$

7. Partial Differential Equations & BVP

Boundary Value Problem using Finite Difference: Replace derivatives in $y'' + p(x)y' + q(x)y = r(x)$ with:

$$\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} + p_i \frac{y_{i+1} - y_{i-1}}{2h} + q_i y_i = r_i$$

Solve Laplace Equation ($\nabla^2 u = 0$): Using 5-point formula:

$$u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1} - 4u_{i,j} = 0$$

Solve Poisson Equation ($\nabla^2 u = f(x, y)$):

$$u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1} - 4u_{i,j} = h^2 f(x_i, y_j)$$